

Starting 7:05

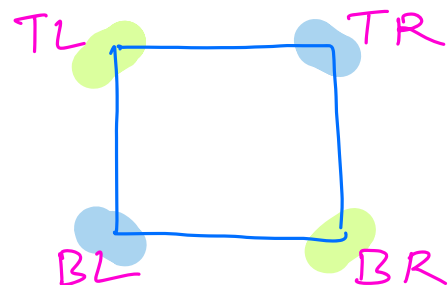
Today's Agenda:

- ① Sub Matrix Sum Queries
- ② Sum of all sub-matrices.
- ③ Max Sub Matrix Sum
- ④ Find an element in row wise & column-wise matrix.

① Given a matrix of size $N \times M$.
For each query, find sum of given sub-matrix.

Ex:-

	0	1	2	3
0	2	-1	3	2
1	3	2	6	2
2	10	9	8	2
3	4	-1	2	3
4	3	2	6	9



Queries

(T, 2)

(B, R)

(2, 1)

(4, 2) = 26.

(1, 1)

(3, 3) = 33.

∴ TL & BR
OR

BL & TR are
fixed.

a unique
submatrix
is defined.

Idea 1:-

For every query, traverse
the entire submatrix.

T.C $\rightarrow O(n \times m \times q)$

Idea 2:-

Using

Prefix

Sum.

	0	1	2	3	4
0					
1					
2					
3					
4					

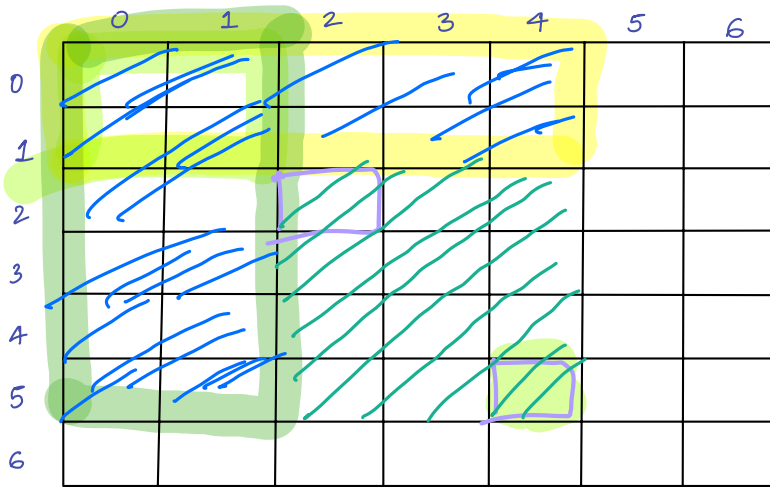
$pSum[i][j]$

$\rightarrow$

sum of all
elements from
(0,0) to (i,j)

$pSum[3][4]$

Assume $\rightarrow$ $pSum[][]$ is built.



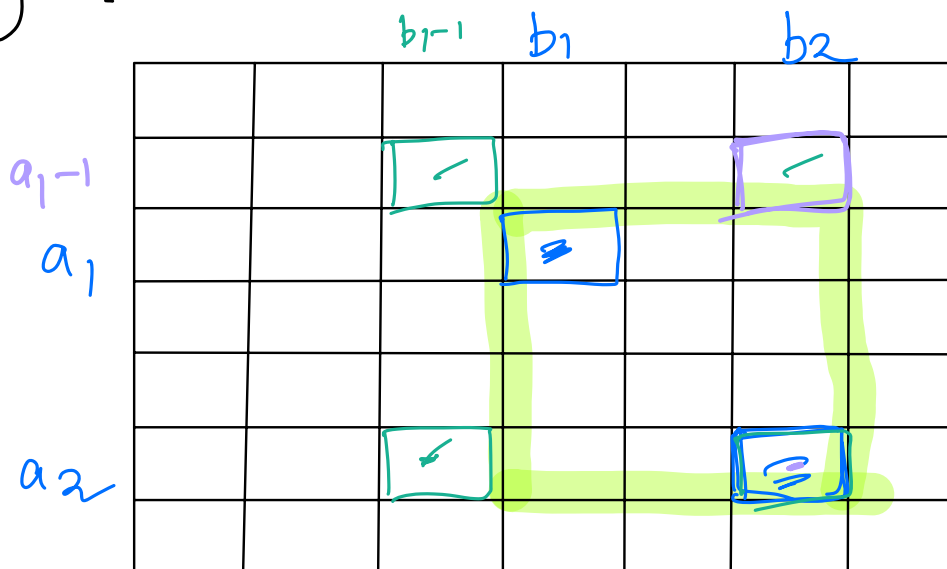
Query

T.L $\rightarrow$ $[2, 2]$

B.R $\rightarrow$ $[5, 4]$

$$pSum[5][4] - pSum[5][1] \\ - pSum[1][4] \\ + pSum[1][1].$$

Generalisation



T.L $\rightarrow$ (a_1, b_1)
B.R $\rightarrow$ (a_2, b_2)

$\rightarrow$ ≥ 0

$$pSum[a_2][b_2] - pSum[a_2][b_1 - 1]$$

$$- \text{pSum}[a_{1-i}][b_2] \\ + \text{pSum}[a_{1-i}][b_{1-i}]$$

	0	1	2
0	a_0	a_1	a_2
1	b_0	b_1	b_2
2	c_0	c_1	c_2

→ Given Matrix.

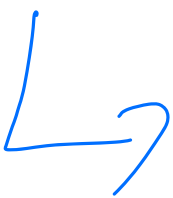
$$\text{pSum}[1][2] \rightarrow a_0 + a_1 + a_2 \\ + b_0 + b_1 + b_2$$

$$\text{pSum}[2][0] \rightarrow a_0 + b_0 + c_0$$

$$\text{pSum}[2][1] \rightarrow a_0 + a_1 \\ + b_0 + b_1 \\ + c_0 + c_1$$

Apply
pSum on
every
row.

$$\begin{bmatrix} a_0 & a_1 + a_0 & a_1 + a_2 + a_0 \\ b_0 & b_1 + b_0 & b_2 + b_1 + b_0 \\ c_0 & c_1 + c_0 & c_2 + c_1 + c_0 \end{bmatrix}$$



Apply pSum on every column

	0	1	2
0	a_0	$a_1 + a_0$	$a_1 + a_2 + a_0$
1	$b_0 + a_0$	$b_1 + b_0 + a_1 + a_0$	$b_2 + b_1 + b_0 + a_2 + a_1 + a_0$
2	$c_0 + b_0 + a_0$	$c_1 + c_0 + b_1 + b_0 + a_1 + a_0$	$c_2 + c_1 + c_0 + b_2 + b_1 + b_0 + a_2 + a_1 + a_0$

$$1 < N, m \leq 10^3$$

$$1 < A[i] \leq 10^5$$

$$10^5 \times (10^3 \times 10^3) = 10^{11}$$

Code.

Algorithm.

- ① On a given matrix, find the row-wise pSum.
- ② On the result from previous

matrix, find the column-wise
pSum.

③

Run a loop over Q
queries & find the
answer using the above
formula.

```
for (i = 0; i < N; i++) {  
    for (j = 0; j < M; j++) {  
        pSum[i][j] = arr[i][j];  
    }  
}
```

```
for (i = 0; i < N; i++) {  
    for (j = 1; j < M; j++) {  
        pSum[i][j] = pSum[i][j-1]  
            + arr[i][j];  
    }  
}
```

for ($j=0; j < m; j++$) {

for ($i=1; i < N; i++$ {

$$pSum[i][j] = pSum[i][j]$$

$$+ pSum[i-1][j];$$

}

Q. 4.

a_1	a_2	b_1	b_2
-	-	-	-

for ($k=0; k < Q; k++$)

{ $a_1 = a[k][0]; a_2 = a[k][1];$
 $b_1 = a[k][2]; b_2 = a[k][3];$

$$ans = pSum[a_2][b_2] - pSum[a_2][b_1-1] \\ - pSum[a_1-1][b_2] + pSum[a_1-1][b_1-1].$$

print(ans);

}

$$T.C = O(Nm + Q)$$

$$S.C = O(N * m).$$

	0	1	2	3
0	a_1	a_2	b_1	b_2
1				
2				
...				
Q				

$Q \times 4$

② - Given a matrix $N \times m$.
Calculate the sum of all submatrix.

	0	1
0	3	1
1	-1	-2
2	2	4

$$4 \times 2 \dots$$

$$2 \times 4 \dots$$



$$[3]$$

$$[3, 1]$$

$$\begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ -1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ -1 & -2 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$[1]$$

$$\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$

$$[-1]$$

$$[-1 \quad -2]$$

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -2 \\ 2 & 4 \end{bmatrix}$$

$$[-2]$$

$$\begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$[2]$$

$$[2, 4]$$

$$[4].$$

① Brute Force Approach.

Fix the T.L. $\rightarrow N \times m$.

$\hookrightarrow$ Fix the B.R.

Q1) $\Rightarrow$ Traverse...

$\downarrow$
 $N \times m$.

$$T.C \rightarrow O(N^2 \times m^2).$$

② Contribution Technique.

	0	1	2	3	4
0	T.L	T.L	T.L		
1	T.L	T.L	T.L		
2	T.L	T.L	T.L B.R.	B.R.	B.R.
3			B.R.	B.R.	B.R.
4			B.R.	B.R.	B.R.

To define a submatrix where a cell is contributing, I need possible values of TC & BR.

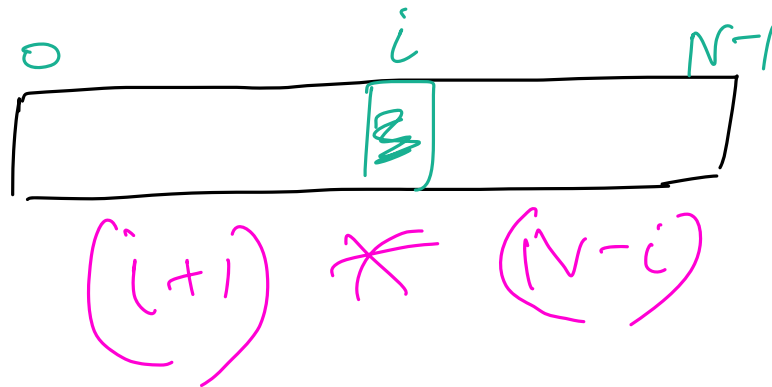
Total no. of matrices in which a certain element will

$$= T.L \times B.R$$

Total
poss.

Total
poss.

$$= 9 \times 9 = 81 \text{ submatrices.}$$



	0	1	2	3	4
0	-	-	-		
1	-	-	-	-	-
2			-	-	-
3			-	-	-

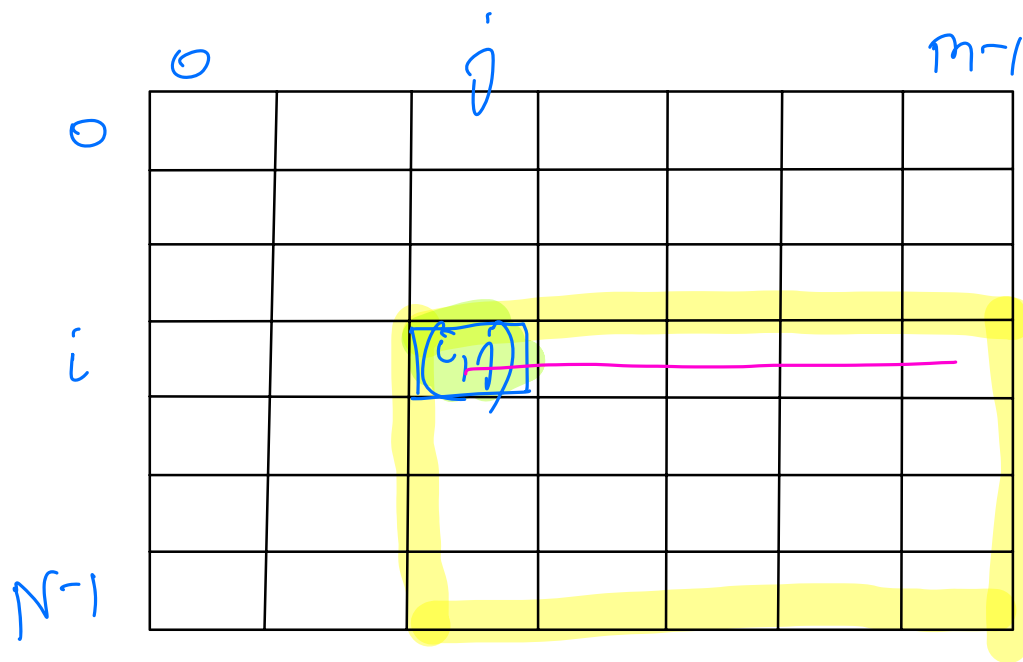
$$T.L = 6.$$
$$B.R = 9.$$

Total.

$$= 9 \times 6$$
$$= 54.$$

Generalise.

Consider a $N \times M$ matrix.



$(1, 2)$

Option for T.L.

$$(j+1) \times (i+1)$$

Option for B.P.

$$(n-i) \times (m-j)$$

$[i, n-1]$

$\rightarrow$

$$n-1-i+1$$

$[j, m-1]$

$\rightarrow$

$$m-1-j+1$$

#

Total contribution of
a cell at $[i][j]$

$$= (j+1) \times (i+1) \times (n-i)$$

$$* (m-j)$$

$$* arr[i][j]$$

#

Code.

```
long arr = 0;
for (i = 0; i < N; i++)
{
    for (j = 0; j < m; j++)
    {
        ans += (
            (j+1) * (i+1) * (n-i)
            * (m-j)
            * arr[i][j]
        );
    }
}
```

Break Untill 9:00 am.

③ Matrix Submatrix Sum.

Given a row-wise & column wise sorted matrix A of size $N \times M$. Find max. submatrix sum.

-10	-6	-2	1
-5	-4	4	5
-3	0	6	20
2	5	10	25

always max.

largest element

Interested in a submatrix ending at last cell. (25).

Idea 1 - Build a suffix sum matrix & find the max. value in that suffix

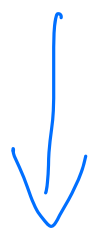
Sum

matrix

Qr

How to build?

$$\begin{bmatrix} -10 & -6 & -2 & 1 \\ -5 & -4 & 4 & 5 \\ -3 & 0 & 6 & 20 \\ 2 & 5 & 10 & 25 \end{bmatrix}$$



Build S.S. Row wise

$$\begin{bmatrix} -17 & -7 & -1 & 1 \\ 0 & 5 & 9 & 5 \\ 23 & 26 & 26 & 20 \\ 42 & 40 & 35 & 25 \end{bmatrix}$$



Apply

S.S. Sum Col.
wise.

$$\begin{bmatrix} 48 & 64 & 69 & 51 \end{bmatrix}$$

65	71	70	50
65	66	61	45
42	40	35	25

Code.

```
long ssum[N][M];
```

```
for (i = 0; i < N; i++)
```

```
{ for (j = 0; j < m; j++)
```

```
{ ssum[i][j] = arr[i][j];
```

```
}
```

```
for (i = N-1; i >= 0; i--)
```

```
{ for (j = m-2; j >= 0; j--)
```

```
{
```

```
ssum[i][j] +=
```

```
ssum[i][j+1];
```

3

```
for ( j = m-1; j >= 0; j--)
{
    for ( i = N-2; i >= 0; i--)
    {
        ssum[i][j] += ssum
        [i+1][j];
    }
}
```

3

long ans = INT_MIN;

```
for ( i = 0; i < N; i++)
{
    for ( j = 0; j < m; j++)
    {
        ans = max ( ans,
        ssum[i][j];
    }
}
```

3

T.C $\rightarrow O(N * m)$.

$$S.C \rightarrow O(N \times m);$$

Q. Given a matrix where every row & every column is sorted. Find an element k .

$N \times m$

	0	1	2	3	4	5
0	-1	2	4	5	9	11
1	1	4	7	8	10	14
2	3	7	9	10	12	18
3	6	10	12	14	16	20
4	9	13	16	19	22	24
5	11	15	19	21	24	27
6	14	20	25	29	31	39
7	17	24	29	32	34	42

$$k = 25$$

Idea 1:

Traverse on each & every element in the matrix.

$$O(N \times m).$$

Idea 2: Apply B.S on every row. & find the answer. $\rightarrow N \log m$

Code

```
i = 0; j = m - 1
while (i < N && j >= 0)
{
    if (arr[i][j] == k)
        { return true; }
    else if (arr[i][j] < k)
        { i++; }
    else if (arr[i][j] > k)
        { j--; }
}
```

$$T.C \rightarrow O(N+M)$$

$$S.C \rightarrow O(1).$$

Done .

Doubts .

Sorted array:

se:
ei

$[-3 \quad -2 \quad 3 \quad 5 \quad 8]$

$[11 \quad 14 \quad 16 \quad 13 \quad 8]$

$\begin{matrix} 0 & 1 & 2 & 3 \\ [1 & 2 & 3 & 4] \end{matrix}$

se = 3. = 6.
end = 2.

1 2 3

1 2 3 4

2 3

2 3 4

3

3 4